

A JOURNEY TOWARD A GREENER, SMARTER FUTURE

Green Skilling, Sustainability & AI



Sustainable Development • Renewable Energy • Artificial Intelligence

Name

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Stream

B.Tech Computer Science

INTRODUCTION

About Me



Raj Lad

B.Tech – Computer Science

Passionate about **Artificial Intelligence** and its role in building a **sustainable future** for our planet.



ENROLLMENT NUMBER

240840131054



BRANCH / STREAM

B.Tech Computer Science



COLLEGE

R.N.G Patel Institute of Technology



SEMESTER

As per current academic year



Why AI & Sustainability?

I am a Computer Science student exploring how technology can protect the environment. I'm keen to build AI-driven solutions — from smart monitoring systems to data-driven decision tools — that support a cleaner, more sustainable world.

Things I Learned



SDG 17

Partnerships & collaboration driving global sustainability goals.



Green Data Centres

Energy-efficient, eco-friendly digital infrastructure.



Govt. Initiatives

National policies and missions promoting a greener India.



Renewable Energy

Solar, wind & clean power replacing fossil fuels.



AI for Environment

Smart applications that monitor and protect nature.

05

Key Learning Areas Explored

SDG 17 – Partnerships for the Goals

17

GLOBAL GOAL

Partnerships for the Goals

Why it's my favorite — it connects every other goal through collaboration.



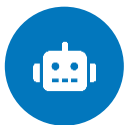
Why It Matters

No single nation, company, or individual can solve climate change, poverty, or pollution alone. SDG 17 unites governments, businesses, and citizens toward shared sustainability targets.



Collaboration in Action

Cross-sector partnerships pool funding, technology, and expertise — accelerating progress on clean energy, education, and equitable growth far faster than isolated efforts.



How AI Strengthens SDG 17

AI enables real-time data sharing, predictive analytics, and transparent monitoring — helping partners coordinate resources and track progress toward inclusive, sustainable development.

Learning from Data Centre Documentaries



Digital Infrastructure & Energy Demand

Every search, stream, and AI query runs on power-hungry servers — making data centres a key focus for sustainable computing.



Why Data Centres Consume Energy

Continuous server operation, cooling systems, and 24/7 uptime demand massive electricity.



Green Data Centres

Eco-friendly facilities using efficient hardware and sustainable design to cut energy waste.



Renewable Energy Usage

Leading data centres now run on solar and wind power instead of fossil fuels.



AI-Based Cooling & Power Optimization

AI predicts cooling needs and adjusts power use in real time, reducing waste.



Carbon Emission Reduction

Smarter infrastructure and clean energy together shrink the carbon footprint of computing.

Learning from Government Websites



Digital India

Nationwide push for e-governance, digital literacy, and connected infrastructure across the country.



Smart Cities Mission

Urban development using technology for efficient energy, water, waste, and transport systems.



Renewable Energy Missions

Programs like the National Solar Mission expanding clean, affordable power generation.



Environmental Conservation

Afforestation, pollution control, and biodiversity protection initiatives nationwide.

Activity Screenshots



CONCEPT

What is Green Skilling?



Definition Training people with the knowledge and technical skills needed for jobs that protect, restore, or sustain the environment — from clean energy to eco-conscious technology.



Why It Matters

Prepares the workforce to meet climate goals, drive green innovation, and reduce environmental impact across every industry.



Career Opportunities

Opens roles in renewable energy, sustainable manufacturing, environmental data science, and green consulting.

EXAMPLES



Renewable
Energy



Electric
Vehicles



Waste
Management



Sustainable
Agriculture



AI Environmental
Solutions

CONCEPT

What is Sustainability?



Definition Meeting the needs of the present without compromising the ability of future generations to meet their own needs.



Environmental

Protecting ecosystems, biodiversity, air, and water for the long term.



Social

Ensuring equity, wellbeing, and quality of life for all communities.



Economic

Building growth that is resource-efficient and viable long-term.

Why It Matters for the Future

Protects resources, climate, and opportunity so future generations inherit a livable, thriving planet.

Everyday Sustainable Practices

Saving energy and water, reducing waste, recycling, and choosing eco-friendly transport and products.

AI – Benefits & Challenges for Sustainability

BENEFITS



Smart agriculture – precision crop & resource management



Climate prediction – forecasting extreme weather events



Waste segregation – automated sorting & recycling



Energy optimization – smarter grids & consumption



Water management – leak detection & conservation



Forest monitoring – satellite-based deforestation alerts

CHALLENGES



High energy consumption of large AI models



Data centre emissions contributing to carbon output



Electronic waste from constant hardware upgrades



Heavy water usage for server cooling systems



Need for responsible, transparent, and ethical AI

EcoVision AI

Intelligent Environmental Monitoring System



Problem Statement

Rising pollution and resource strain make real-time, data-driven environmental monitoring essential for timely, informed action.



Objectives

- Monitor air & environmental quality in real time
- Automate waste classification using computer vision
- Predict pollution trends with machine learning
- Visualize carbon footprint & impact data

TECHNOLOGIES



Python



Machine
Learning



Computer
Vision



IoT



Cloud

KEY FEATURES



Air quality monitoring



Waste classification



Pollution prediction



Carbon footprint dashboard

Expected Impact

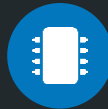
Empowers communities with real-time environmental insight — directly advancing SDG 13 (Climate Action) and SDG 11 (Sustainable Cities).



Thank You

Questions & Discussion

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